

End-of-Chapter Review Exercise 1 — Full Solutions

Questions 1–3 (complete)

Past-paper questions: substitution to transform roots, and power sums via the recurrence relation

Colour key: *green = verification check* | *blue = alternative method*.

Question 1

Paper 1 Q7, November 2010

The roots of $x^3 + 4x - 1 = 0$ are α, β, γ . Using $y = \frac{1}{1+x}$, show that $6y^3 - 7y^2 + 3y - 1 = 0$ has roots $\frac{1}{\alpha+1}, \frac{1}{\beta+1}, \frac{1}{\gamma+1}$. For $n = 1, 2$ find $\sum \frac{1}{(\alpha+1)^n}$, deduce the case $n = 3$, and hence evaluate $\sum \frac{(\beta+1)(\gamma+1)}{(\alpha+1)^2}$.

The transformed equation. From $y = \frac{1}{1+x}$ we get $1+x = \frac{1}{y}$, so $x = \frac{1-y}{y}$. Substituting into $x^3 + 4x - 1 = 0$ and multiplying by y^3 :

$$(1-y)^3 + 4(1-y)y^2 - y^3 = 0.$$

Expanding, $(1 - 3y + 3y^2 - y^3) + (4y^2 - 4y^3) - y^3 = -6y^3 + 7y^2 - 3y + 1 = 0$, i.e.

$$6y^3 - 7y^2 + 3y - 1 = 0.$$

Each root is $y = \frac{1}{1+x}$, so the roots are $\frac{1}{\alpha+1}, \frac{1}{\beta+1}, \frac{1}{\gamma+1}$, as required.

Write $T_n = \sum \frac{1}{(\alpha+1)^n}$ for the power sums of this cubic ($a = 6, b = -7, c = 3, d = -1$), with

$$\sum \frac{1}{\alpha+1} = -\frac{b}{a} = \frac{7}{6}, \quad \sum \frac{1}{(\alpha+1)(\beta+1)} = \frac{c}{a} = \frac{1}{2}, \quad \prod \frac{1}{\alpha+1} = -\frac{d}{a} = \frac{1}{6}.$$

$n = 1, 2$: $T_1 = \sum \frac{1}{\alpha+1} = \boxed{\frac{7}{6}}$, and using the squaring identity

$$T_2 = \left(\sum \frac{1}{\alpha+1}\right)^2 - 2 \sum \frac{1}{(\alpha+1)(\beta+1)} = \left(\frac{7}{6}\right)^2 - 2\left(\frac{1}{2}\right) = \frac{49}{36} - 1 = \boxed{\frac{13}{36}}.$$

Deduce $n = 3$. Multiplying each root's equation $6y^3 - 7y^2 + 3y - 1 = 0$ by y^n and summing over the three roots gives the recurrence

$$6T_{n+3} - 7T_{n+2} + 3T_{n+1} - T_n = 0, \quad T_0 = 3.$$

At $n = 0$ (leading term $6T_3$):

$$6T_3 - 7T_2 + 3T_1 - T_0 = 0 \implies 6T_3 = 7\left(\frac{13}{36}\right) - 3\left(\frac{7}{6}\right) + 3 = \frac{73}{36} \implies T_3 = \boxed{\frac{73}{216}}.$$

Hence the final sum. Put $p = \alpha + 1, q = \beta + 1, r = \gamma + 1$, so $pqr = \frac{1}{1/6} = 6$. Each term satisfies $\frac{qr}{p^2} = \frac{pqr}{p^3} = \frac{6}{p^3}$, so

$$\sum \frac{(\beta+1)(\gamma+1)}{(\alpha+1)^2} = 6 \sum \frac{1}{(\alpha+1)^3} = 6T_3 = 6 \cdot \frac{73}{216} = \boxed{\frac{73}{36}}. \quad \blacksquare$$

Check. Numerically $T_1 = 1.1667 = \frac{7}{6}$, $T_2 = 0.3611 = \frac{13}{36}$, $T_3 = 0.3380 = \frac{73}{216}$, and the target sum evaluates to $2.0278 = \frac{73}{36} \checkmark$.

Question 2

Paper 11 Q11, November 2014

The roots of $x^4 + 4x^3 + 2x^2 - 4x + 1 = 0$ are $\alpha, \beta, \gamma, \delta$. Find (i) $\sum \alpha$, (ii) $\sum \alpha^2$, (iii) $\sum \frac{1}{\alpha}$, (iv) $\sum \frac{\alpha}{\beta\gamma\delta}$. Then, using $y = x + 1$, form and solve a quartic in y and hence find the roots.

Here $a = 1$, $b = 4$, $c = 2$, $d = -4$, $e = 1$, so by Vieta's formulae

$$\sum \alpha = -4, \quad \sum \alpha\beta = 2, \quad \sum \alpha\beta\gamma = 4, \quad \alpha\beta\gamma\delta = 1.$$

(i) $\sum \alpha = \boxed{-4}$.

(ii) $\sum \alpha^2 = \left(\sum \alpha\right)^2 - 2\sum \alpha\beta = 16 - 4 = \boxed{12}$.

(iii) $\sum \frac{1}{\alpha} = \frac{\sum \alpha\beta\gamma}{\alpha\beta\gamma\delta} = \frac{4}{1} = \boxed{4}$.

(iv) $\frac{\alpha}{\beta\gamma\delta} = \frac{\alpha^2}{\alpha\beta\gamma\delta} = \alpha^2$, so $\sum \frac{\alpha}{\beta\gamma\delta} = \frac{\sum \alpha^2}{\alpha\beta\gamma\delta} = \frac{12}{1} = \boxed{12}$.

Substitution $y = x + 1$ (so $x = y - 1$). Substituting and expanding,

$$(y - 1)^4 + 4(y - 1)^3 + 2(y - 1)^2 - 4(y - 1) + 1 = 0.$$

The y^3 and y coefficients vanish and the constant is 4:

$$y^4 - 4y^2 + 4 = 0, \quad \text{i.e.} \quad (y^2 - 2)^2 = 0.$$

So $y^2 = 2$ (a double root), giving $y = \pm\sqrt{2}$ (each repeated). Since $x = y - 1$,

$$x = -1 + \sqrt{2} \text{ (double)}, \quad x = -1 - \sqrt{2} \text{ (double)}.$$

Check. The double roots mean $x^4 + 4x^3 + 2x^2 - 4x + 1 = (x^2 + 2x - 1)^2$; expanding the right side reproduces the quartic $\checkmark$. Numerically $\sum \alpha = -4$, $\sum \alpha^2 = 12$, $\sum \frac{1}{\alpha} = 4 \checkmark$.

Question 3

Paper 13 Q8, June 2012

The cubic $x^3 - x^2 - 3x - 10 = 0$ has roots α, β, γ . (i) With $u = -\alpha + \beta + \gamma$, show $u + 2\alpha = 1$ and find the cubic with roots $-\alpha + \beta + \gamma$, $\alpha - \beta + \gamma$, $\alpha + \beta - \gamma$. (ii) State $\alpha\beta\gamma$ and find the cubic with roots $\frac{1}{\beta\gamma}, \frac{1}{\gamma\alpha}, \frac{1}{\alpha\beta}$.

By Vieta's formulae ($a = 1$, $b = -1$, $c = -3$, $d = -10$):

$$\sum \alpha = 1, \quad \sum \alpha\beta = -3, \quad \alpha\beta\gamma = 10.$$

(i) Since $\alpha + \beta + \gamma = 1$,

$$u = -\alpha + \beta + \gamma = (\alpha + \beta + \gamma) - 2\alpha = 1 - 2\alpha, \quad \text{so} \quad u + 2\alpha = 1.$$

The three target roots are $1 - 2\alpha$, $1 - 2\beta$, $1 - 2\gamma$, so each new root u comes from a root x of the cubic via $u = 1 - 2x$, i.e. $x = \frac{1-u}{2}$. Substitute this directly into $x^3 - x^2 - 3x - 10 = 0$:

$$\left(\frac{1-u}{2}\right)^3 - \left(\frac{1-u}{2}\right)^2 - 3\left(\frac{1-u}{2}\right) - 10 = 0.$$

Multiplying through by 8 to clear denominators:

$$(1 - u)^3 - 2(1 - u)^2 - 12(1 - u) - 80 = 0.$$

Expanding each bracket,

$$(1 - 3u + 3u^2 - u^3) - 2(1 - 2u + u^2) - 12(1 - u) - 80 = -u^3 + u^2 + 13u - 93 = 0,$$

so, multiplying by -1 ,

$$\boxed{u^3 - u^2 - 13u + 93 = 0}.$$

Alternative — symmetric functions of $1 - 2\alpha$. Working directly with the new roots,

$$\sum (1 - 2\alpha) = 3 - 2 \sum \alpha = 3 - 2(1) = 1,$$

$$\sum (1 - 2\alpha)(1 - 2\beta) = 3 - 2(2 \sum \alpha) + 4 \sum \alpha\beta = 3 - 4 - 12 = -13,$$

$$\prod (1 - 2\alpha) = 1 - 2 \sum \alpha + 4 \sum \alpha\beta - 8 \alpha\beta\gamma = 1 - 2 - 12 - 80 = -93.$$

These give $u^3 - (1)u^2 + (-13)u - (-93) = u^3 - u^2 - 13u + 93 = 0$, the same cubic.

(ii) $\alpha\beta\gamma = \boxed{10}$. Now $\frac{1}{\beta\gamma} = \frac{\alpha}{\alpha\beta\gamma} = \frac{\alpha}{10}$, and likewise for the others, so the target roots are $\frac{\alpha}{10}, \frac{\beta}{10}, \frac{\gamma}{10}$. Substitute $y = \frac{x}{10}$, i.e. $x = 10y$:

$$(10y)^3 - (10y)^2 - 3(10y) - 10 = 0 \implies 1000y^3 - 100y^2 - 30y - 10 = 0,$$

and dividing by 10,

$$\boxed{100y^3 - 10y^2 - 3y - 1 = 0}.$$

Check. (i) The roots $1 - 2\alpha$ give sum 1, pair-sum -13 , product -93 ✓. (ii) The roots $\frac{\alpha}{10}$ give sum $\frac{1}{10}$, pair-sum $-\frac{3}{100}$, product $\frac{1}{100}$, i.e. $100y^3 - 10y^2 - 3y - 1 = 0$ ✓.